Automatic injection Molding

1. Objective:- To study about Automatic Injection machine and Produce 10 articles by using automatic injection molding at optimize cycle time and calculate the Production rate.

2. Machine/eqt./Accessories/Tools required:-
Automatic injection moulding machine, injection mould, stop watch, Weighing machine, Pyrometer, brass rod, vernier caliper, micrometer, Plastic raw material.

3. Principle:- A moulding Procedure whereby a heat-softened Plastic material is forced from a cylinder into a relatively cool cavity giving the article the desired shape.

4. Procedure:- 1. Firstly, we have to clean the machine with cotton cloth.
2. Arrange all the required tools.
3. On the mains switch of the machine.
4. Now set the barrel temperature according to raw material.
5. Plastic raw material is stored in the hopper.
6. The Plastic falls on the rotating screw.
7. It is conveyed forward by the flights of the screw and is gradually melted as it is moved forward.
8. When enough melted Plastic has accumulated ahead of the screw, the screw stops rotating.
9. The screw acting as the Plunger Pushes the molten Plastic out of the injection barrel.
10. The Plastic goes through the nozzle, sprue, runner, gates and into the mould cavities.

11. Packing occurs and little more plastic is forced in after the cavities are full.
12. Packing continues till the gate freezes.
13. When the plastic in the gates freezes the pressure on the screw releases the screw rotates back to melt and conveys the next shot of plastic ahead of the screw.
14. Parts are cooled enough to hold their shape, the mould opens the parts are ejected.
15. Next cycles will perform automatically.

5. Result & conclusion:-

Production rate per hour = 3600 x No of cavity / Avg. cycle time in seconds.

We have produced 10 product by using AIMM, Calculated production rate/hour is - 164 prod/hr.

6. Safety:-

1. Don't touch hot barrel.
2. Wear safety goggles & gloves.
3. Follow strictly DO'S & don'ts.
4. Allow soak in period.
5. Safety shield must in proper position when purging out the material from the barrel.
6. Do not temper with the machine safety devices.

7. Precautions:-

1. Use correct grade of material.
2. Ensure the temperature controller is working properly.
3. Dry material if required.
4. Ensure proper lubrication of movable parts.
5. Ensure mould centering.

Expt. No. _____ Date _____

6. Ensure proper cooling.

Machine specification

Injection unit		Clamping unit	
Screw dia. mm	50	Clamping force Ton	100
Screw speed rpm	181	Mould hight min/max. mm	150/320
Injection Rate cm³/s	99	opening stroke, mm	360
Max Inj. capacity cm³	512	Total day light max. mm	360
Max. Injection Pressure bar	140	Distance between tie bars hor x vert. mm	360 x 360
Injection speed	99		
Refilling speed	181	Platen size, mm	530 x 530
Nozzle contact Pressure Tons		Day cycle Time - sec/stroke mm	
		Rotary Drive Power kW	

Observation/calculation:

S. No	Mould closing time (se)	Injection time (sec)	Hold on time (sec)	Cooling time (sec)	Mould opening time (sec)	Ejection time (sec)	Total cycle time (sec)
1.	3.45	2.76	0.50	13.11	1.76	0.82	22.4
2.	3.45	2.76	0.50	13.17	1.76	0.82	22.46
3	3.45	2.76	0.49	13.08	1.76	0.83	22.34
4.	3.44	2.75	0.50	12.88	1.76	0.83	22.16
5.	3.44	2.76	0.50	12.67	1.76	0.82	21.95
6.	3.44	2.76	0.50	12.18	1.76	0.82	21.46
7	3.44	2.76	0.50	11.54	1.76	0.81	20.81

→ cycle incomplete

Teacher's Signature _____

Expt. No. Date

Moulding defects observed

S. No.	Short Shot	Flash	voids	Sink marks	Weld line	Burn marks	Cracks	Splay	Jetting
1st Shot	✓								
2nd shot	✓				✓				
3rd shot	✓				✓				
4th Shot	✓								
5th Shot		✓			✓				
6th Shot			✓						
7th Shot		✓							
8th Shot			✓						
9th Shot				✓					
10th Shot									

Teacher's Signature

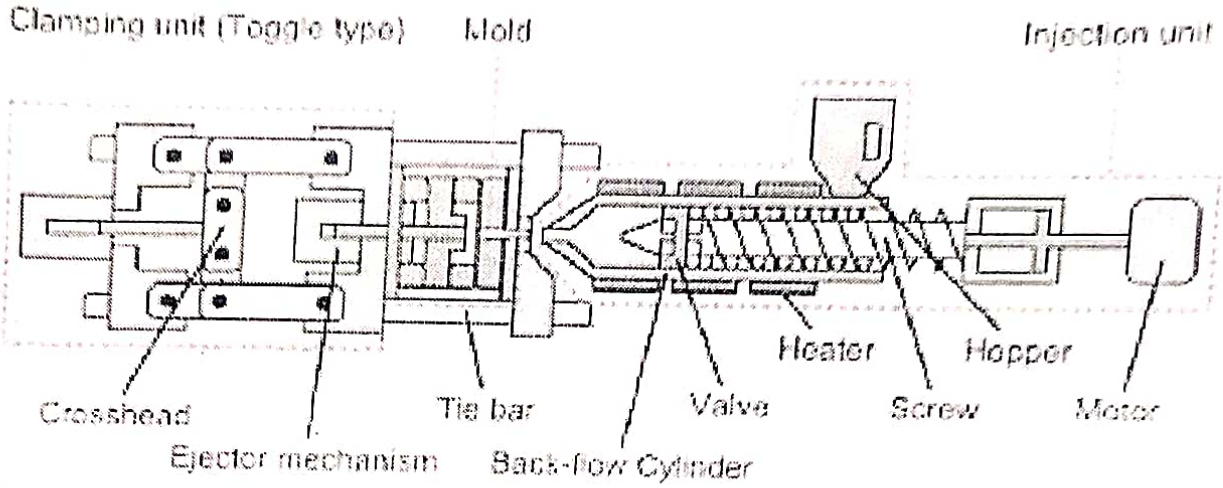
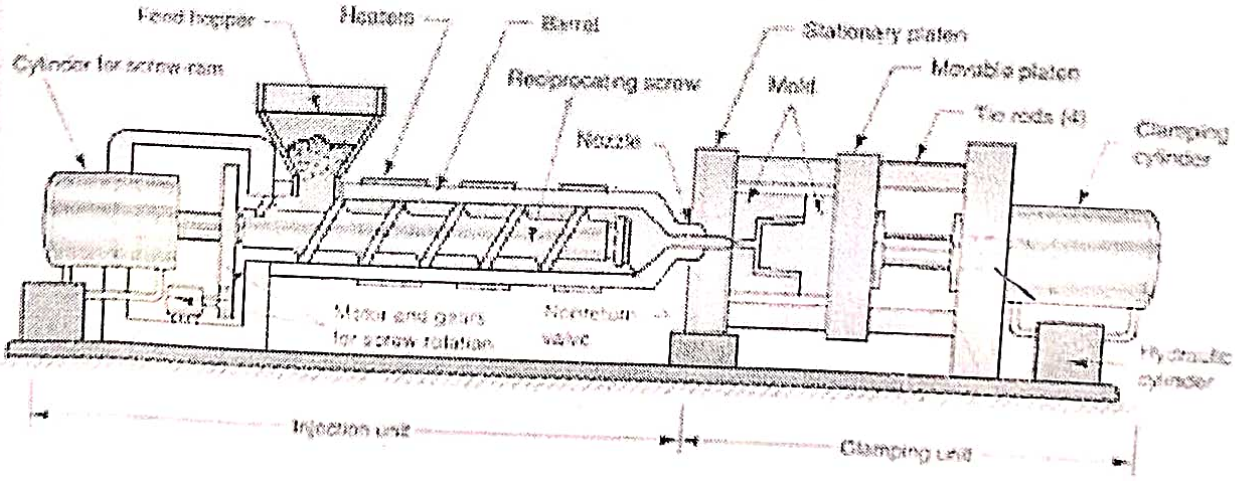
Average Cycle time = 22.4 + 22.46 + 22.34 + 22.16 + 21.95 + 21.46 + 20.81

$$\Rightarrow \frac{153.58}{7} \Rightarrow 21.94 \text{ (sec)}$$

$$\text{Production Rate Per hour} = \frac{3600 \times \text{No. of cavity}}{\text{Average Cycle time}}$$

$$\Rightarrow \frac{3600 \times 1}{21.94} \Rightarrow \frac{3600}{21.94} \Rightarrow 164 \text{ Product Per hour.}$$

INJECTION MOULDING



Parts of Injection Moulding Machine